



There are plans to take the GPU outside the cabinet, tethered to the PC via a PCIe x16 connector

If you've been reading *Digit* religiously (is there any other way to read it?), you would be well aware of the changes that graphics cards are going through (for some background on where GPUs are headed, read *A New DirectXion*, *Digit*, October 2006). In a nutshell, GPUs are becoming CPU-like; so much so that a day when you see a GPU performing CPU tasks and vice-versa is not very far away. What's more, you might very well see the two integrate—imagine a central processor with bits of GPU connected to it, much like the design of the PlayStation 3 console.

Graphic cards are perhaps the most exciting frontier in the coming years: even as NVIDIA and ATI release DirectX 10 products with as much as 1 GB onboard memory (yikes!), Intel is rumoured to be working secretly on its own GPU design, taking a more traditional x86 route—rumours talk about a CPU-GPU hybrid (a CGPU?) code-named Larrabee, consisting of sixteen cores. The cores are small, in-order x86 mini-processors with a short pipeline and lots of vector hardware to implement some GPU-oriented extension of the x86 instruction set. AMD, too, has talked of its x86-GPU plans, aka Fusion, which plans to marry AMD processors with tech acquired from ATI GPUs. Integrating a processor and a graphics unit like this has many advantages, the biggest

Graphics cards

Big, hot, and crazy

being that of the flexibility possible: tasks pertaining to shaders, vector math, CPU-intensive physics, sound, and procedural art and animation could all be balanced on the same chip, making for an extremely efficient piece of silicon. Moreover, a GPU based on the x86 instruction set can leverage years of programming experience, tools and techniques allowing for lowered software development costs.

As things stand, graphics cards are expensive. DirectX 10 cards are currently only available from NVIDIA, and as of this writing, only in two variants—"will financially ruin you" and "will financially ruin your entire family." With competition, prices will fall, and ATI's DX10 monster has just been spotted in the wild.

Prices aren't the only means of putting the hurt on you, though: that ATI card, for example is a nine-inch behemoth with its memory running at slightly more than 1 GHz and a core speed of about 800 MHz. These DirectX 10 cards pull in more than 200 watts, and it wouldn't be a stretch to see vapour- or water-cooling units becoming the norm rather than the exception when the ATI parts appear. And this is just when the cards are running in single mode—throw in SLI and step back, way back: the power requirements would be astronomical (just today we spotted a Chinese PSU on the Net rated at 1000 W!).

There are also plans to take the GPU outside the cabinet, tethered to the PC via a PCIe x16 connector. This will not only prevent total meltdown of your system but would also offer fast graphics to, say, a laptop, or allow for a scenario in which banks of external graphic processors

are shared over a network by several computers.

As for what's currently on tap for future buyers—NVIDIA mostly, with ATI to release its line of DX10 cards between March and May '07. NVIDIA's GeForce 8 series looks like this: the 8800 GTX and GTS variants, which vary slightly in clock speeds, and the GTS coming in a leaner 320 MB version, which is also much more affordable than its 640 MB sibling. Also planned is an 8900 GTX core with 25 per cent more shader power (clocked at 700 MHz GPU and 2200 MHz memory). An SLI-on-a-card variant of this is also planned: the GeForce 8950 GX2 will be a dual-chip card based on a new 80-nm G80 chip, clocked at 550 MHz with 2x512MB of 256-bit GDDR4 (memory at 2000 MHz).

ATI's DX10 cards are rumoured to be 65-nm parts, meaning they'll run cooler. The DX10 R600 and RV600 series from ATI will come in various guises—high-end, low-end, dual-cores on a card, and even the external cards mentioned above (code-named Lasso). The R600XTX is the top-of-the-line card, recently spotted with a massive four-heatpipe-powered cooler. It comes with 1024 MB of GDDR4 memory running at over 2000 MHz; 12 inches long, the card is said to consume more than 240 W.

Apart from pushing pixels, these new cards are also HD-ready: ready to decode the highly CPU-taxing codecs of generation next—MPEG-2, H.264, and Microsoft's VC1 format. They also offer full HDCP support—over either DVI-D or HDMI. To be future-proof in the coming months, ensure that any monitors you purchase offer either HDCP over DVI-D, or come with an HDMI port.

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